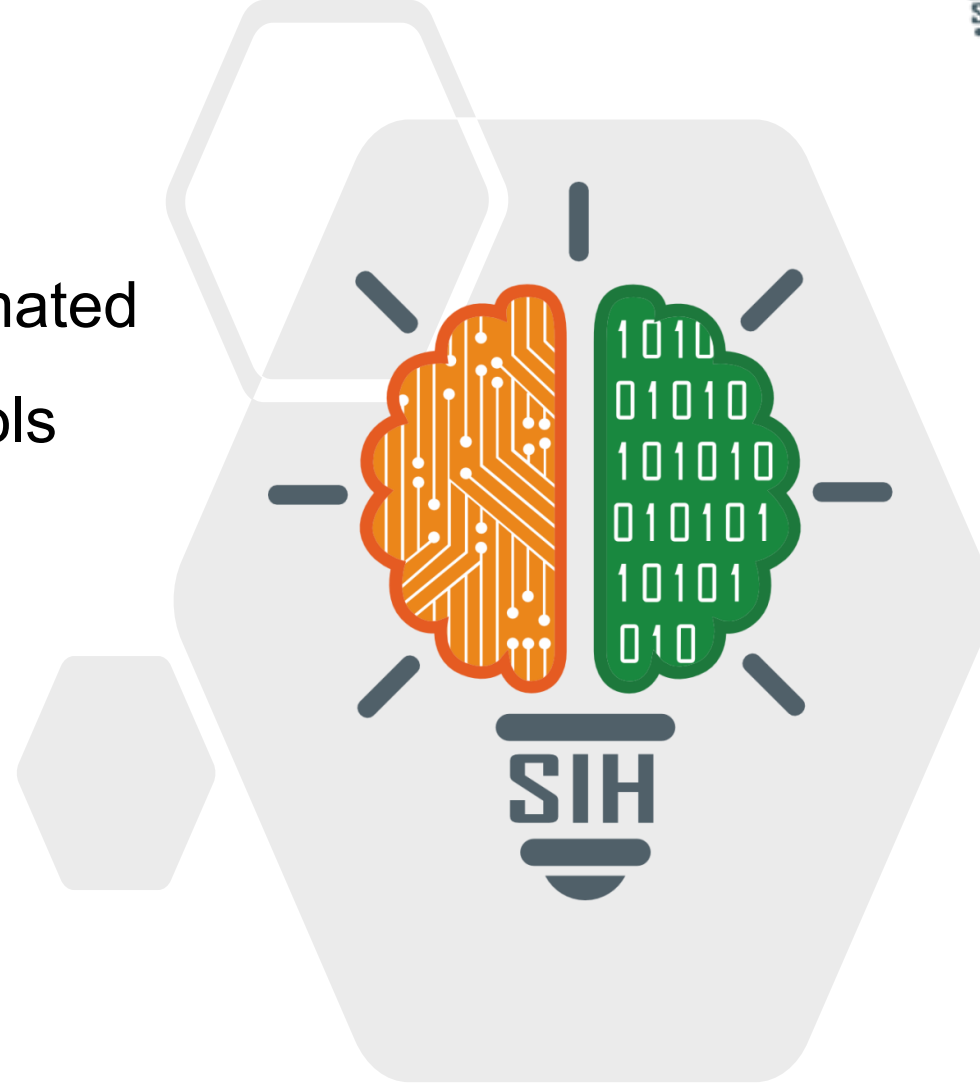


SMART INDIA HACKATHON 2025



- **Problem Statement ID:** 25012
- **Problem Statement Title:** Automated Attendance System for Rural Schools
- **Theme:** Smart Education
- **PS Category:** Software
- **Team ID:** 61692
- **Team Name:** StrategIQ



Problem Statement

Rural schools often use **manual attendance systems**, which are **slow**, **error-prone**, and reduce **teaching time**. These lead to **inaccurate records**, affecting **government schemes** like mid-day meals and weakening **student tracking**.

Innovation at the Core

- **Offline** compatibility for low connectivity zones.
- **Proxy Proof** dual authentication system with hybrid RFID + Face recognition.
- **Multilingual support** for beneficiaries of rural areas.
- **Fast deployment** made easy as it requires **no previous infrastructure** from schools.
- Use of AI model on cloud to determine trends for implementation of schemes from Education Board.

Our Solution

StrategIQ's Digital Hazri: An Intelligent Attendance Companion

- **Core Concept:** Automate and secure attendance using an IoT-driven **RFID + face recognition system**.
- **Seamless Attendance Workflow:** RFID scan triggers the ESP32-CAM to capture an image, verify it against the stored face data, and confirm student presence in real-time.
- Dedicated **Web Dashboard for 4** main targeted audience: Teachers, Students/Parents, School Administrator and Government Bodies.
- Provides class-wise attendance, daily reports, and instant absentee lists **aligned with the school timetable**.

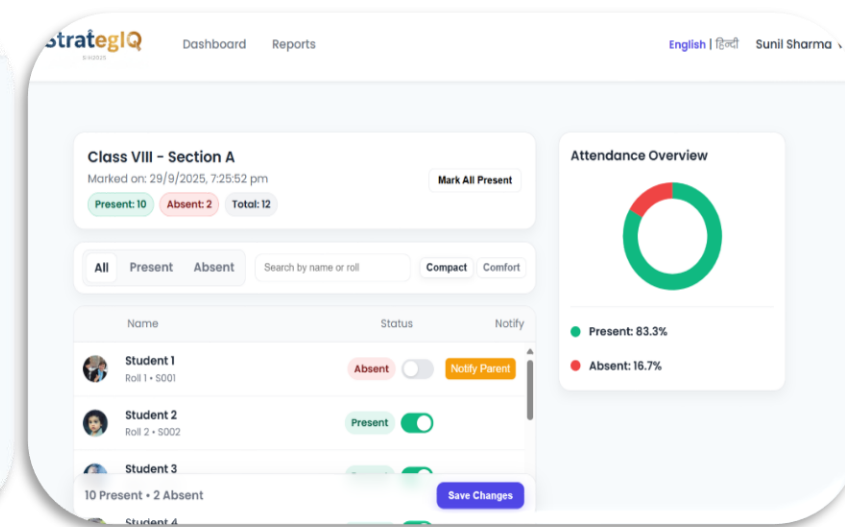
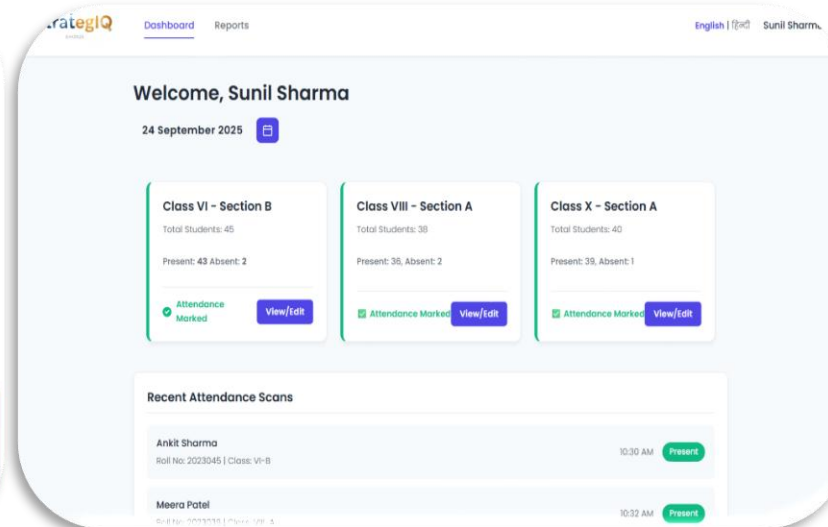
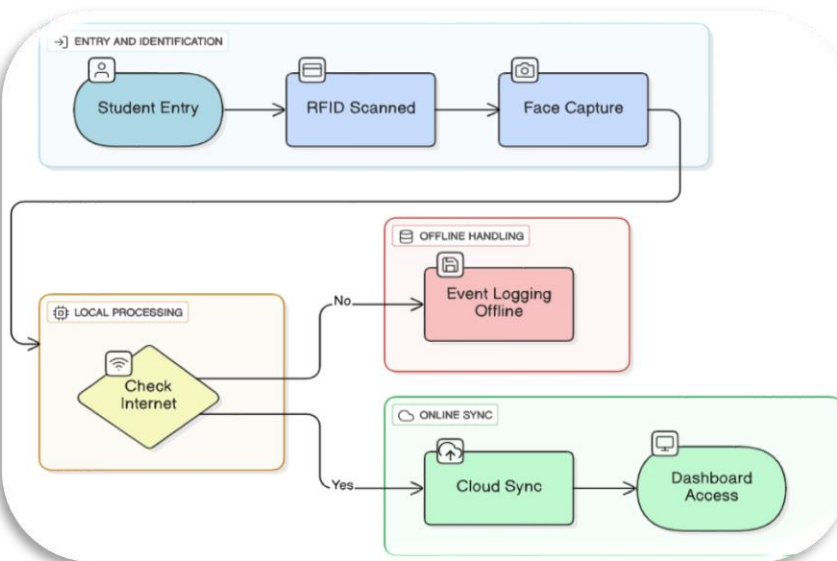
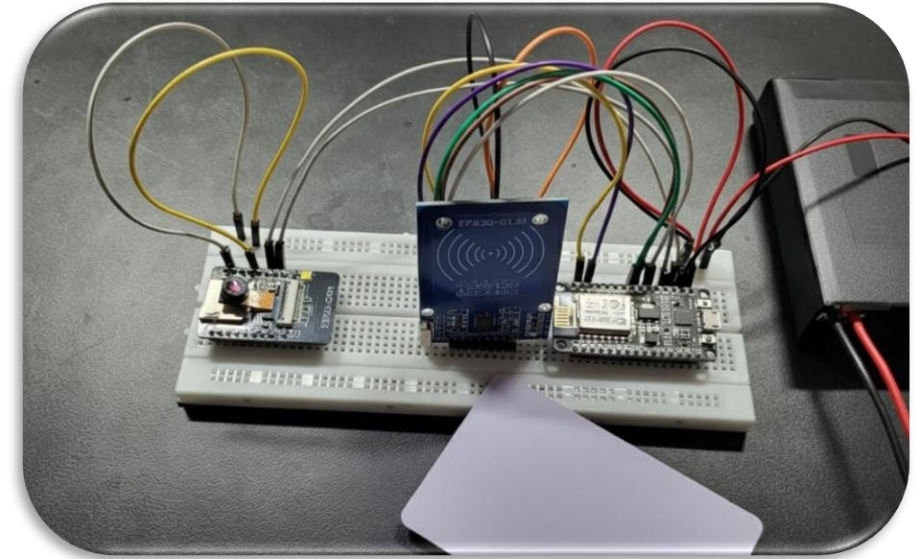
Tech Stack Used

Hardware: ESP32-CAM, ESP8266 + MFRC522 (RFID)

Languages: Embedded C (firmware), JavaScript (dashboard & Firebase functions),

Cloud: Firebase Firestore (attendance logs),
Firebase Storage (images), Firebase Auth (secure access)

Frameworks/Tools: React.js (dashboard),
ESP-WHO (lightweight face recognition), REST APIs



Feasibility:

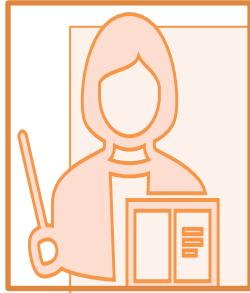
- **Low cost, easily available hardware** with each node costing around **800-900 Rs** making large scale deployment practical.
- **Multilingual support** ensures that people in **rural areas** with less English exposure can operate it efficiently.
- **Offline logging** ensures that records are maintained even in **low connectivity zones** of states.
- **Minimal Dashboard design** focusing only on attendance for **ease of use** of teachers and parents.

Challenges:

- ESP-WHO works well in controlled lighting, but **accuracy may drop in poor lighting** or crowded entries.
- **Per scan time may increase** due to on device processing of face recognition due to low processing power.
- **Increased cost** of Firebase cloud services for actual deployment.

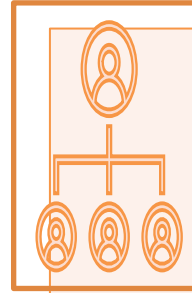
Future Strategies to Overcome Challenges:

- Using **better quality camera** for actual deployment of system.
- A **local computer** may be later connected **for processing** which decreases per scan time significantly.
- **Local server** may be used instead of using Firebase cutting cloud costs **OR** purchasing **Enterprise plan** of Firebase by State Education Board.



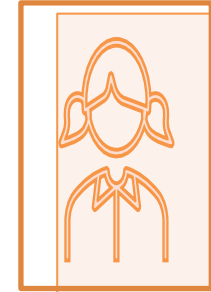
Teachers

Saves 10-15 minutes per class, removing any errors and makes checking previous records easier



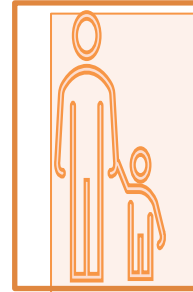
Administration

Centralized Dashboard with minimal infrastructure and reduces staff workload.



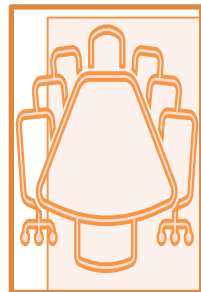
Students

Fair attendance system discouraging immoral behaviour and have records of attendance for future purpose



Parents

Easily able to check if student has reached school or not through web dashboard ensuring student safety



Education Board

Accurate data for schemes, policy insights and easy to scale for rural schools.

- **ASER 2024 National findings**
Pratham Non Profit Organization
[First Section II.pmd](#)
- **Biometric Attendance in Schools**
Ministry of Education, Govt. of India – *Initiatives on Digital Attendance*
<https://www.education.gov.in/>
- **Face Recognition on ESP32-CAM**
Espressif Official Documentation – *ESP32-CAM AI Thinker Guide*
<https://docs.espressif.com/projects/esp-idf/en/latest/esp32/esp32-cam/>
- **RFID-based Attendance Systems**
Research Paper: *RFID Based Smart Attendance Monitoring System* – IEEE Xplore
<https://ieeexplore.ieee.org/document/8468057>
- **Firebase for Real-time Databases**
Firebase Official Documentation – *Realtime Database & Authentication*
<https://firebase.google.com/docs>
- **Hybrid IoT + AI in Education**
World Bank Report on *Technology in Education*
<https://documents.worldbank.org/en/publication/documentreports/documentdetail/964121500168465630/technology-in-education>